Canonical Zeckendorf Normalization and the Minimal Berstel Adder

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Abstract

We study the canonical normalization map behind addition in Zeckendorf numeration. Digitwise superposition of two admissible Zeckendorf words yields a finitely supported word over $\{0, 1, 2\}$, and addition is recovered by replacing this overlay with the unique admissible word of the same Fibonacci value. We realize this normalization as the canonical section of a Fibonacci congruence and describe it by a terminating local rewrite system whose irreducible configurations are exactly the admissible Zeckendorf words. We then separate this one-sided low-to-high canonical normalizer from the classical most-significant-digit-first adders of Frougny and Berstel. In the one-sided model, the canonical map has coordinatewise unbounded anticipation, is not subsequential, and has exact prefix-destruction index $\Delta(n) = n$. By contrast, the classical Berstel transducer belongs to a different subsequential setting, and we prove that its state complexity is exactly 10. Finally, we show that every fixed-radius synchronous local realization of canonical normalization requires a number of rounds growing linearly with the input scale.

Keywords. Zeckendorf numeration, canonical normalization, subsequential transducer, Berstel adder, online delay, Fibonacci addition.

Mathematics Subject Classification. 68Q45, 68Q70, 11B39.

1 Background and Main Results

Zeckendorf addition can be viewed as a normalization problem. If $c, d \in X_\infty$ are admissible Zeckendorf words, then their digitwise overlay $c + d$ is a finitely supported word over $\{0, 1, 2\}$, and the arithmetic task is to replace this noncanonical word by the unique admissible word with the same Fibonacci value. Thus the central object of the paper is the canonical normalization map

$$\mathrm{Fold}_\infty : \{0, 1, 2\}^{(\mathbb{N})} \to X_\infty.$$

Our purpose is to describe this map in structural terms and to compare two different finite-state viewpoints on Fibonacci addition.

Finite-state arithmetic in nonstandard numeration systems is classical, especially in Pisot bases [**?**, **?**, **?**]. Recent work continues to treat Fibonacci-based numeration from an automata-theoretic viewpoint; see, for instance, [**?**, **?**]. For broader finite-state normalization phenomena in linear and Ostrowski-type numeration systems, see also [**?**, **?**]. The present note remains strictly Fibonacci-specific and does not attempt PLRS, k-bonacci, or multidimensional generalizations. The point

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of departure here is that the canonical map Fold_∞ on one-sided low-to-high streams should not be conflated with the classical most-significant-digit-first adders. We therefore begin with the normalization mechanism itself. First, we realize Fold_∞ as the canonical section of the Fibonacci congruence on the free commutative monoid of scale counts and give a terminating local rewrite system whose irreducible configurations are exactly the admissible Zeckendorf words. This yields the factorization

$$\text{superpose digits} \longrightarrow \mathrm{Fold}_\infty$$

for Zeckendorf addition.

The first complexity statement is negative, but its novelty is task-specific rather than historical. Earlier work of Frougny already showed that finite-state behavior in Fibonacci numeration depends delicately on the direction of reading and on the chosen conversion problem [**?**, **?**, **?**]. In particular, finite-state normalization results in the literature often pass through compositions of one-sided subsequential transformations rather than a single canonical low-to-high map [**?**]. Our negative result concerns exactly that single canonical map. What we add here is a sharp formulation for the canonical one-sided normalization map itself. In the low-to-high model natural for Fold_∞, every fixed output coordinate has unbounded anticipation. Equivalently, the map admits no finite one-pass online delay. More strongly, once finitely supported inputs are encoded as low-to-high finite words, the resulting normalization function is not subsequential. The obstruction is not a minor technicality: for every fixed lookahead bound, there are two inputs agreeing on an arbitrarily long prefix whose selected normalized digits nevertheless differ, and one may even choose a prefix pair for which the first normalized digit flips. In fact, for every prescribed output length n there is a prefix-related input pair whose normalized outputs disagree from the very first symbol while the shorter output has length exactly n. This sharply separates canonical one-sided normalization from the classical online-addition constructions of Frougny, where the input is read most significant digit first and the output is not the greedy expansion [**?**, Prop. 13]. The witnesses may already be chosen inside the genuine addition image $\{c + d : c, d \in X_\infty\}$.

We then revisit the classical 10-state Berstel adder in its proper setting. The existence of this transducer, and a 10-state presentation of it, are classical [**?**, **?**]. Our new point is an exact minimality statement. Using deterministic subsequential Nerode equivalence, we show that this classical transducer is already minimal: every deterministic subsequential realization of the same recoding requires at least 10 reachable states.

This sequential distinction should be compared with the literature on parallel addition in nonstandard numeration systems [**?**]. Our final result shows that the canonical map Fold_∞ remains intrinsically nonlocal in the synchronous setting as well: every fixed-radius local normalization scheme requires a number of rounds growing linearly with the input scale.

The paper is organized as follows. Section 2 introduces the quotient description of Zeckendorf normalization. Section 3 gives the rewrite system and identifies addition with overlay plus normalization. Section 4 proves that the one-sided map Fold_∞ has coordinatewise unbounded anticipation, no bounded online delay, and no subsequential realization in the low-to-high finite-word model. Sections 5 and 6 turn to the classical Berstel adder and prove that its deterministic subsequential state complexity is exactly 10. Section 7 contains the local parallel lower bound.

2 Canonical Normalization in Zeckendorf Numeration

2.1 Admissible Words and the Value Map

Fibonacci weights. Throughout the paper we use the convention

$$F_1 = F_2 = 1, \qquad F_{n+2} = F_{n+1} + F_n \quad (n \geq 1).$$

The weight at position k is F_{k+1}.

Admissible representatives. Let

$$X_\infty := \left\{ c = (c_1, c_2, \ldots) \in \{0,1\}^{\mathbb{N}} : \mathrm{supp}(c) \text{ is finite and } c_k c_{k+1} = 0 \text{ for all } k \right\}.$$

For $c \in X_\infty$ we define

$$\mathrm{Val}(c) := \sum_{k \geq 1} c_k F_{k+1} \in \mathbb{N}.$$

By Zeckendorf's theorem, every $N \in \mathbb{N}$ admits a unique representative $Z(N) \in X_\infty$ such that $\mathrm{Val}(Z(N)) = N$ [**?**].

Free scale counts. Let

$$\mathcal{D} := \{a = (a_1, a_2, \ldots) : a_k \in \mathbb{N},\ \mathrm{supp}(a) \text{ finite}\},$$

with pointwise addition. We write e_k for the kth standard basis vector.

Definition 2.1 (Fibonacci congruence)**.** Let $\equiv$ be the least commutative-monoid congruence on $\mathcal{D}$ generated by

$$e_2 \equiv 2e_1 \qquad \text{and} \qquad e_{k+2} \equiv e_{k+1} + e_k \quad (k \geq 1).$$

Definition 2.2 (Value map on scale counts)**.** For $a \in \mathcal{D}$ we define

$$\mathrm{Val}(a) := \sum_{k \geq 1} a_k F_{k+1}.$$

Proposition 2.3 (Completeness of Fibonacci congruence)**.** *If $a \equiv b$ in $\mathcal{D}$, then* $\mathrm{Val}(a) = \mathrm{Val}(b)$*. Consequently the value map descends to a well-defined monoid homomorphism*

$$\overline{\mathrm{Val}} : \mathcal{D}/{\equiv} \to \mathbb{N}.$$

Proof. It is enough to check the defining relations. For the recurrence relation,

$$\mathrm{Val}(e_{k+2}) = F_{k+3} = F_{k+2} + F_{k+1} = \mathrm{Val}(e_{k+1} + e_k).$$

For the base relation,

$$\mathrm{Val}(e_2) = F_3 = 2 = 2F_2 = \mathrm{Val}(2e_1).$$

The claim follows because $\equiv$ is generated from these relations by additivity. □

Theorem 2.4 (The quotient monoid is $\mathbb{N}$)**.** *The homomorphism $\overline{\mathrm{Val}} : \mathcal{D}/{\equiv} \to \mathbb{N}$ is an isomorphism of commutative monoids. Moreover every congruence class contains a unique representative in X_∞.*

Proof. Surjectivity is immediate from Zeckendorf's theorem: for every $N \in \mathbb{N}$, $Z(N) \in X_\infty$ satisfies $\mathrm{Val}(Z(N)) = N$. For injectivity, write $\varepsilon = [e_1] \in \mathcal{D}/\equiv$. By induction on the recurrence relation,

$$[e_k] = F_{k+1}\varepsilon \qquad (k \geq 1).$$

Hence every class $[a] \in \mathcal{D}/\equiv$ satisfies

$$[a] = \sum_{k\geq 1} a_k[e_k] = \left(\sum_{k\geq 1} a_k F_{k+1}\right)\varepsilon = \mathrm{Val}(a)\varepsilon.$$

So the congruence class is uniquely determined by its value. The uniqueness of the representative in X_∞ is then exactly the uniqueness of the Zeckendorf expansion. □

2.2 Canonical Representatives and the Fold Section

Corollary 2.5 (The fold map is a computable section)**.** *The embedding $X_\infty \hookrightarrow \mathcal{D}$ is a computable section of the quotient map $\mathcal{D} \twoheadrightarrow \mathcal{D}/\equiv \cong \mathbb{N}$. Equivalently, the map*

$$\mathrm{Fold}_\infty : \mathcal{D} \to X_\infty, \qquad \mathrm{Fold}_\infty(a) := Z(\mathrm{Val}(a)),$$

is a well-defined normalization map selecting the unique admissible representative in each congruence class.

Proof. The statement is a restatement of Theorem 2.4. □

3 Overlay and Normalization

3.1 Local Value-Preserving Rewrites

Although Fold_∞ is defined quotient-theoretically, its action can be described directly by local digit rewrites on $\mathcal{D}$. For $a = (a_1, a_2, \ldots) \in \mathcal{D}$, consider the following directed moves:

(R1)	$a \to a - e_k - e_{k+1} + e_{k+2}$	if $a_k \geq 1,\ a_{k+1} \geq 1$,
(R2)	$a \to a - 2e_1 + e_2$	if $a_1 \geq 2$,
(R3)	$a \to a - 2e_2 + e_1 + e_3$	if $a_2 \geq 2$,
(R4)	$a \to a - 2e_k + e_{k-2} + e_{k+1}$	if $k \geq 3,\ a_k \geq 2$.

Each rule preserves value, since it is a directed form of one of the identities

$$F_{k+1} + F_{k+2} = F_{k+3}, \qquad 2F_2 = F_3, \qquad 2F_3 = F_2 + F_4, \qquad 2F_{k+1} = F_{k-1} + F_{k+2}\ (k \geq 3).$$

Proposition 3.1 (Termination and unique normal form)**.** *Starting from any $a \in \mathcal{D}$, every rewrite sequence generated by* (R1)–(R4) *terminates. The irreducible configurations are exactly the admissible words X_∞, and the terminal irreducible word is* $\mathrm{Fold}_\infty(a)$.

Proof. Define

$$A(a) := \sum_{k\geq 1} a_k, \qquad B(a) := \sum_{k\geq 1} k\, a_k, \qquad C(a) := a_2,$$

and compare triples $\mu(a) := (A(a), B(a), C(a))$ lexicographically. Rule (R1) and rule (R2) decrease A by 1. Rule (R4) preserves A and decreases B by 1. Rule (R3) preserves both A and B, but

decreases C by 2. Hence every rewrite step strictly decreases $\mu(a)$, so infinite rewrite sequences are impossible.

If a is irreducible, then no adjacent pair of positive digits can occur, or else (R1) would apply; likewise no digit can be ≥ 2, or else one of (R2)–(R4) would apply. Thus every irreducible configuration is a finitely supported $\{0,1\}$-word with no adjacent 1s, i.e. an element of X_∞. Conversely, no rule applies to an element of X_∞, so the irreducibles are exactly X_∞.

Every rewrite step preserves Val, hence the terminal irreducible word has the same value as a. By Corollary 2.5, there is a unique admissible word with this value, namely $\mathrm{Fold}_\infty(a)$. □

Remark 3.2. Proposition 3.1 does not assert confluence of the abstract rewrite system generated by (R1)–(R4). What is used here is weaker and sufficient for the paper: every rewrite sequence terminates, every terminal irreducible is admissible, every rewrite preserves value, and each value class contains a unique admissible representative.

For overlays of two admissible inputs the initial alphabet is only $\{0,1,2\}$, but it is convenient to formulate the rewrite system on the whole monoid $\mathcal{D}$. This avoids case distinctions and keeps the normalization target equal to the canonical section from Corollary 2.5.

3.2 Stable Addition

For $c, d \in X_\infty$ we define the digitwise superposition

$$(c+d)_k := c_k + d_k \in \{0,1,2\}.$$

Since c and d have finite support, $c+d$ can be viewed as an element of $\mathcal{D}$.

Definition 3.3 (Stable addition)**.** For $c, d \in X_\infty$, define

$$c \oplus d := Z(\mathrm{Val}(c) + \mathrm{Val}(d)).$$

The previous subsection shows that $\oplus$ can be computed without leaving the digit system.

Theorem 3.4 (Addition is superposition followed by normalization)**.** *For all $c, d \in X_\infty$ one has*

$$c \oplus d = \mathrm{Fold}_\infty(c+d).$$

Equivalently, Zeckendorf addition is the normalization of the noncanonical digitwise overlay. Thus the addition problem reduces to the study of the canonical map Fold_∞ *on the input alphabet* $\{0,1,2\}$*; the remainder of the paper compares its one-sided sequential behavior with the classical most-significant-digit-first Berstel adder.*

Proof. By definition of Fold_∞,

$$\mathrm{Fold}_\infty(c+d) = Z(\mathrm{Val}(c+d)).$$

Since Val is additive on finitely supported digit strings,

$$\mathrm{Val}(c+d) = \mathrm{Val}(c) + \mathrm{Val}(d).$$

Therefore

$$\mathrm{Fold}_\infty(c+d) = Z(\mathrm{Val}(c) + \mathrm{Val}(d)) = c \oplus d.$$

□

Once the canonical section has been fixed, the arithmetic problem reduces to realizing Fold_∞ by a finite streaming device with bounded delay [**?**, **?**, **?**].

Example 3.5 (A short normalization trace)**.** Using the low-to-high Fibonacci-scale convention, let

$$c = (1, 0, 1, 0, 0), \qquad d = (1, 0, 0, 1, 0) \in X_\infty.$$

Then

$$c + d = (2, 0, 1, 1, 0),$$

so the overlay contains both a digit 2 and the forbidden pattern 11. Its value is

$$\mathrm{Val}(c + d) = 2F_2 + F_4 + F_5 = 2 + 3 + 5 = 10 = F_3 + F_6.$$

Therefore

$$\mathrm{Fold}_\infty(c + d) = Z(10) = (0, 1, 0, 0, 1) = c \oplus d.$$

This is the smallest type of example in which normalization must resolve the two standard obstructions simultaneously.

4 Unbounded Anticipation and Non-Subsequentiality

4.1 One-Sided Online Delay

Definition 4.1 (One-pass online delay)**.** Let $\mathcal{A}_{\mathrm{in}}$ and $\mathcal{A}_{\mathrm{out}}$ be finite alphabets. A map

$$T : \mathcal{A}_{\mathrm{in}}^{(\mathbb{N})} \to \mathcal{A}_{\mathrm{out}}^{(\mathbb{N})}$$

has *one-pass online delay* $\delta \in \mathbb{N}$ if, for every $j \geq 1$, the output digit $(T(x))_j$ depends only on the input prefix $x_1, \ldots, x_{j+\delta}$. Equivalently, after reading the first n input digits, a sequential device with delay δ has already determined the first $\max\{n - \delta, 0\}$ output digits.

Remark 4.2. Definition 4.1 is the one-sided low-to-high notion adapted to Fold_∞ on finitely supported streams. It is not the most-significant-digit-first anticipation parameter used in the classical online-addition literature [**?**]. This distinction is essential.

Theorem 4.3 (Coordinatewise unbounded anticipation)**.** *For every output coordinate $j \geq 1$ and every $\delta \geq 0$, there exist finitely supported inputs $x, y \in \{0, 1, 2\}^{(\mathbb{N})}$ such that*

$$x_i = y_i \qquad (1 \leq i \leq j + \delta),$$

but

$$(\mathrm{Fold}_\infty(x))_j \neq (\mathrm{Fold}_\infty(y))_j.$$

Proof. Fix $j \geq 1$ and $\delta \geq 0$.

If $j = 1$, choose $m \geq 1$ with $2m \geq \delta + 1$ and define

$$x := \sum_{t=0}^{m-1} e_{2t+1}, \qquad y := x + 2e_{2m+1}.$$

The two inputs agree on the first $\delta + 1$ positions. Since x is already admissible, $\mathrm{Fold}_\infty(x) = x$, so $(\mathrm{Fold}_\infty(x))_1 = 1$. Moreover

$$\mathrm{Val}(x) = \sum_{t=1}^{m} F_{2t} = F_{2m+1} - 1,$$

hence

$$\mathrm{Val}(y) = F_{2m+1} - 1 + 2F_{2m+2} = F_{2m+4} - 1 = \sum_{t=1}^{m+1} F_{2t+1}.$$

Therefore

$$\mathrm{Fold}_\infty(y) = \sum_{t=1}^{m+1} e_{2t},$$

and so $(\mathrm{Fold}_\infty(y))_1 = 0$.

Assume now that $j \geq 2$. Choose $m \geq 1$ with $2m - 1 \geq \delta$ and set

$$x := \sum_{t=0}^{m-1} e_{j+2t}, \qquad y := x + 2e_{j+2m}.$$

The two inputs agree on the first $j + 2m - 1$ positions, hence on the first $j + \delta$ positions. Again x is admissible, so $\mathrm{Fold}_\infty(x) = x$ and $(\mathrm{Fold}_\infty(x))_j = 1$.

For the second input, one computes

$$\mathrm{Val}(y) = \sum_{t=0}^{m-1} F_{j+2t+1} + 2F_{j+2m+1} = F_{j-1} + \sum_{t=0}^{m} F_{j+2t+2}.$$

The last identity may be checked directly by induction on m from the Fibonacci recurrence. Hence the admissible word

$$z := Z(F_{j-1}) + \sum_{t=0}^{m} e_{j+1+2t}$$

has the same value as y. By uniqueness of admissible representatives, $\mathrm{Fold}_\infty(y) = z$. Since all nonzero digits of the second summand lie strictly to the right of position j, and $Z(F_{j-1})$ is supported in positions $< j$, one has $z_j = 0$. Thus $(\mathrm{Fold}_\infty(y))_j = 0$. □

Corollary 4.4 (No finite one-pass delay)**.** *The canonical normalization map* Fold_∞ *admits no finite one-pass online delay.*

Proof. If Fold_∞ had one-pass online delay δ, then the jth output digit would be determined by the first $j + \delta$ input digits for every $j \geq 1$. This contradicts Theorem 4.3. □

Corollary 4.5 (The obstruction already occurs on genuine addition inputs)**.** *For every output coordinate $j \geq 1$ and every $\delta \geq 0$, there exist admissible words $c, d, c', d' \in X_\infty$ such that the overlays $c + d$ and $c' + d'$ agree on the first $j + \delta$ coordinates, but*

$$\big(c \oplus d\big)_j \neq \big(c' \oplus d'\big)_j.$$

Proof. In the proof of Theorem 4.3, the word x is always admissible, so it can be written as $x = x + 0$ with $x, 0 \in X_\infty$.

If $j = 1$, the second witness is

$$y = \sum_{t=0}^{m-1} e_{2t+1} + 2e_{2m+1} = \left(\sum_{t=0}^{m} e_{2t+1}\right) + e_{2m+1},$$

and both summands are admissible.

If $j \geq 2$, the second witness is

$$y = \sum_{t=0}^{m-1} e_{j+2t} + 2e_{j+2m} = \left(\sum_{t=0}^{m} e_{j+2t} \right) + e_{j+2m},$$

and again both summands are admissible. Hence the witness family already lies in the image of genuine digitwise superposition of admissible inputs. The conclusion then follows from Theorem 3.4. □

4.2 A Finite-Word Consequence

To compare the one-sided canonical map with standard transducer models, we encode finitely supported inputs as low-to-high finite words. For $u = u_1 \cdots u_n \in \{0,1,2\}^*$, write

$$\langle u \rangle := \sum_{i=1}^{n} u_i e_i \in \mathcal{D}.$$

If $a \in \{0,1\}^{(\mathbb{N})}$ is finitely supported, let $\mathrm{trim}(a)$ denote the finite word obtained by deleting trailing zeros. We then define

$$\mathcal{N}(u) := \mathrm{trim}(\mathrm{Fold}_\infty(\langle u \rangle)) \qquad (u \in \{0,1,2\}^*).$$

Lemma 4.6 (Bounded terminal discrepancy for subsequential transducers)**.** *Let $T : \Sigma^* \to \Gamma^*$ be realized by a deterministic subsequential transducer with terminal output map ω. Set*

$$L := \max_q |\omega(q)|.$$

If u is a prefix of v, then $T(u)$ and $T(v)$ have a common prefix of length at least $|T(u)| - L$.

Proof. Write $v = uw$. Let α be the output produced while reading u, and let q be the state reached after u. Then

$$T(u) = \alpha\, \omega(q).$$

If the run is continued along w, the output has the form

$$T(v) = \alpha\, \beta\, \omega(q')$$

for some word β and some terminal state q'. Thus $T(u)$ and $T(v)$ share the prefix α, whose length is

$$|\alpha| = |T(u)| - |\omega(q)| \geq |T(u)| - L.$$

□

Theorem 4.7 (The one-sided canonical normalizer is not subsequential)**.** *The finite-word normalization map*

$$\mathcal{N} : \{0,1,2\}^* \to \{0,1\}^*$$

is not realizable by any deterministic subsequential transducer reading the input from low Fibonacci scales to high Fibonacci scales.

Proof. Suppose, for contradiction, that a deterministic subsequential transducer realizes $\mathcal{N}$, and let L be the constant from Lemma 4.6. For each $m \geq 1$, consider the prefix pair

$$u_m := (10)^{m-1}1, \qquad v_m := u_m 02.$$

Then

$$\langle u_m \rangle = \sum_{t=0}^{m-1} e_{2t+1}, \qquad \langle v_m \rangle = \sum_{t=0}^{m-1} e_{2t+1} + 2e_{2m+1},$$

so these are exactly the $j = 1$ witnesses from the proof of Theorem 4.3. Hence

$$\mathcal{N}(u_m) = u_m, \qquad \mathcal{N}(v_m) = 0(10)^m 1.$$

In particular, $\mathcal{N}(u_m)$ and $\mathcal{N}(v_m)$ differ in their first symbol, so their common prefix has length 0. On the other hand, u_m is a prefix of v_m, and

$$|\mathcal{N}(u_m)| = |u_m| = 2m - 1.$$

Lemma 4.6 therefore yields

$$0 \geq |\mathcal{N}(u_m)| - L = 2m - 1 - L$$

for every m, which is impossible for $m > L$. This contradiction shows that $\mathcal{N}$ is not subsequential. □

Theorem 4.8 (Exact prefix-destruction index)**.** *For $n \geq 1$, define*

$$\Delta(n) := \max\Big\{ n - \mathrm{lcp}(\mathcal{N}(u), \mathcal{N}(v)) : u \text{ is a proper prefix of } v,\ |\mathcal{N}(u)| = n \Big\}.$$

Then

$$\Delta(n) = n \qquad (n \geq 1).$$

Proof. Since $\mathrm{lcp}(\mathcal{N}(u), \mathcal{N}(v)) \geq 0$, one always has $\Delta(n) \leq n$.

If $n = 2m - 1$ is odd, use the prefix pair

$$u_m = (10)^{m-1}1, \qquad v_m = u_m 02$$

from the proof of Theorem 4.7. There

$$\mathcal{N}(u_m) = u_m, \qquad \mathcal{N}(v_m) = 0(10)^m 1,$$

so $|\mathcal{N}(u_m)| = 2m - 1 = n$ and $\mathrm{lcp}(\mathcal{N}(u_m), \mathcal{N}(v_m)) = 0$. Hence $\Delta(2m - 1) = 2m - 1$.

If $n = 2m$ is even, set

$$a_m := (10)^{m-1}2, \qquad b_m := a_m 1.$$

Then a_m is a proper prefix of b_m. Moreover

$$\mathrm{Val}(a_m) = \sum_{t=1}^{m-1} F_{2t} + 2F_{2m} = (F_{2m-1} - 1) + 2F_{2m} = F_{2m+2} - 1 = \sum_{t=1}^{m} F_{2t+1},$$

so

$$\mathcal{N}(a_m) = (01)^m.$$

Likewise,

$$\mathrm{Val}(b_m) = \mathrm{Val}(a_m) + F_{2m+1} = F_{2m+3} - 1 = \sum_{t=1}^{m+1} F_{2t},$$

and therefore

$$\mathcal{N}(b_m) = 1(01)^m.$$

Thus $|\mathcal{N}(a_m)| = 2m = n$ and $\mathrm{lcp}(\mathcal{N}(a_m), \mathcal{N}(b_m)) = 0$, so $\Delta(2m) = 2m$.

Both parity classes attain the trivial upper bound, hence $\Delta(n) = n$ for all $n \geq 1$. □

Remark 4.9. The point of Theorem 4.7 is not to claim a first non-sequentiality phenomenon in Fibonacci numeration. Rather, it gives a sharpened statement for the specific canonical low-to-high normalization map: the obstruction is coordinatewise, it is witnessed by an explicit family of prefix pairs, and it already appears on genuine addition overlays.

Theorem 4.7 shows that the one-sided map Fold_∞ is not the same object as the classical online adders of Frougny and Berstel. In Frougny's most-significant-digit-first model, addition in base φ is computable with delay 3, and the associated finite-word recoding is subsequential, but the output is not the greedy expansion [**?**, Prop. 13]. The next two sections revisit that classical transducer on its own terms.

5 The Classical Berstel Adder

5.1 A Deterministic Subsequential Transducer

We now switch to the classical finite-word, most-significant-digit-first setting of [**?**, **?**]. The relevant device is the 10-state Berstel adder. Because the final output depends on the last state, the natural model is a deterministic *subsequential* transducer rather than a letter-to-letter Mealy machine.

Its state set is

$$Q = \{000, 001, 002, 010, 100, 101, 0\bar{1}2, 1\bar{1}2, 01\bar{1}, 11\bar{1}\}.$$

The transitions are

$$\begin{aligned}
&000 \xrightarrow{d/0} 00d, \qquad 100 \xrightarrow{d/1} 00d \quad (d = 0, 1, 2),\\
&001 \xrightarrow{0/0} 010, \ \xrightarrow{1/0} 100, \ \xrightarrow{2/0} 101,\\
&002 \xrightarrow{0/0} 11\bar{1}, \ \xrightarrow{1/1} 000, \ \xrightarrow{2/1} 001,\\
&010 \xrightarrow{0/0} 100, \ \xrightarrow{1/0} 101, \ \xrightarrow{2/1} 0\bar{1}2,\\
&101 \xrightarrow{0/1} 010, \ \xrightarrow{1/1} 100, \ \xrightarrow{2/1} 101,\\
&0\bar{1}2 \xrightarrow{0/0} 01\bar{1}, \ \xrightarrow{1/0} 010, \ \xrightarrow{2/0} 100,\\
&1\bar{1}2 \xrightarrow{0/1} 01\bar{1}, \ \xrightarrow{1/1} 010, \ \xrightarrow{2/1} 100,\\
&01\bar{1} \xrightarrow{0/0} 001, \ \xrightarrow{1/0} 002, \ \xrightarrow{2/0} 1\bar{1}2,\\
&11\bar{1} \xrightarrow{0/1} 001, \ \xrightarrow{1/1} 002, \ \xrightarrow{2/1} 1\bar{1}2.
\end{aligned}$$

The terminal output map is

$$\omega(000) = 000,\ \omega(001) = 001,\ \omega(002) = 010,\ \omega(010) = 010,$$

$$\omega(100) = 100,\ \omega(101) = 101,\ \omega(0\bar{1}2) = 000,\ \omega(1\bar{1}2) = 100,$$

$$\omega(01\bar{1}) = 001, \qquad \omega(11\bar{1}) = 101.$$

Let

$$\mathcal{K} = (Q, \delta, \eta, \omega)$$

denote the deterministic subsequential transducer defined above, with initial state 000. For a state $q \in Q$, let

$$F_q : \{0,1,2\}^* \to \{0,1\}^*$$

be the function obtained by starting in q, following the transition labels, and appending the terminal word ω of the last state. For a finite word $w = d_1 \cdots d_n$ read most significant digit first, we write

$$\mathrm{Val}_{\mathrm{MSD}}(w) := \sum_{i=1}^{n} d_i F_{n-i+2}.$$

When discussing residual functions and shortest separating words, we keep the complete output word, including leading zeros, because those zeros belong to the subsequential output. They are immaterial only at the level of the represented Fibonacci value.

All ten displayed states are reachable from the initial state. One may use

$$\varepsilon,\ 1,\ 2,\ 10,\ 11,\ 12,\ 20,\ 102,\ 202,\ 1020$$

to reach respectively

$$000,\ 001,\ 002,\ 010,\ 100,\ 101,\ 11\bar{1},\ 0\bar{1}2,\ 1\bar{1}2,\ 01\bar{1}.$$

Proposition 5.1 (Classical correctness)**.** *The transducer $\mathcal{K}$ is the classical Berstel adder in the most-significant-digit-first Fibonacci setting. It computes a value-preserving binary recoding of words over $\{0,1,2\}$: for every input word w, the complete output $\mathcal{K}(w)$ is binary and satisfies*

$$\mathrm{Val}_{\mathrm{MSD}}(\mathcal{K}(w)) = \mathrm{Val}_{\mathrm{MSD}}(w).$$

In general this output is not the greedy Zeckendorf expansion.

Proof. This is the classical construction; see [**?**, Prop. 13] and [**?**, Thm. 3.1]. Those references show that the machine realizes a value-preserving recoding from ternary Fibonacci input words to binary Fibonacci output words in the most-significant-digit-first convention. The non-greedy statement is already part of Frougny's proposition. □

Example 5.2 (A short non-greedy output witness)**.** Starting from the initial state 000, the input word 121 produces the complete output word 001100. After suppressing the two leading zeros, the represented binary word is 1100, which still contains the forbidden factor 11. Hence the classical Berstel output is not the greedy Zeckendorf expansion of the same value.

Remark 5.3. Theorem 4.4 and Proposition 5.1 concern different sequential functions. The former concerns the canonical one-sided map Fold_∞, whereas the latter concerns the classical Berstel recoding in the most-significant-digit-first model. Example 5.2 exhibits the difference explicitly: the Berstel machine returns a non-greedy value-preserving recoding, while the canonical map of Section 4 is defined to return the greedy representative.

6 Minimality of the Berstel Kernel

6.1 Residual Functions

Define an equivalence relation on Q by

$$q \sim q' \iff F_q = F_{q'}.$$

For deterministic subsequential transducers, this is the usual sequential Nerode equivalence. Any deterministic subsequential transducer implementing the same function as $\mathcal{K}$ must have at least $|Q/\sim|$ reachable states.

To avoid ambiguity, we distinguish the displayed realization $\mathcal{K}$ from the subsequential function that it induces. The reachability statement above shows that $\mathcal{K}$ really has ten reachable states as a concrete machine. The minimality problem addressed below is a statement about the induced function: namely, whether any deterministic subsequential realization of that same function can use fewer than ten reachable states.

Lemma 6.1 (Separation by terminal output)**.** *If $\omega(q) \neq \omega(q')$, then $q \not\sim q'$. Consequently the only pairs not already separated by the empty input are*

$$(000, 0\bar{1}2), \quad (001, 01\bar{1}), \quad (002, 010), \quad (100, 1\bar{1}2), \quad (101, 11\bar{1}).$$

Proof. If $\omega(q) \neq \omega(q')$, then

$$F_q(\varepsilon) = \omega(q) \neq \omega(q') = F_{q'}(\varepsilon),$$

so $q \not\sim q'$. Reading the terminal output list above, one sees that the only equal-terminal pairs are exactly the five pairs displayed. □

Lemma 6.2 (The remaining pairs are separated by the input 0)**.** *The five pairs listed in Lemma 6.1 are all inequivalent. More precisely,*

$$\begin{aligned}
F_{000}(0) &= 0000 \neq 0001 = F_{0\bar{1}2}(0),\\
F_{001}(0) &= 0010 \neq 0001 = F_{01\bar{1}}(0),\\
F_{002}(0) &= 0101 \neq 0100 = F_{010}(0),\\
F_{100}(0) &= 1000 \neq 1001 = F_{1\bar{1}2}(0),\\
F_{101}(0) &= 1010 \neq 1001 = F_{11\bar{1}}(0).
\end{aligned}$$

Proof. Each identity is obtained by following one transition on the input 0 and then appending the terminal output of the arrival state. □

6.2 Minimality of the Classical Kernel

Theorem 6.3 (Minimality of the Berstel kernel)**.** *All states in Q are reachable and pairwise inequivalent under subsequential Nerode equivalence. Consequently the function induced by $\mathcal{K}$ has deterministic subsequential state complexity* 10*, and $\mathcal{K}$ is a minimal realization of that function.*

Proof. For deterministic subsequential transducers, two states are equivalent if and only if every input word produces the same complete output word from both states, including the terminal output. Thus the theorem reduces to separating the residual functions F_q.

By Lemma 6.1, every pair with distinct terminal output is already separated by the empty input. Lemma 6.2 handles the only five remaining pairs. Hence all ten states are pairwise inequivalent, and in fact every inequivalent pair is separated by an input word of length at most 1. Equivalently, the partition-refinement computation reaches the discrete partition

$$\boxed{|Q/\sim| = 10, \qquad \max_{q \neq q'} |w_{q,q'}| \leq 1,}$$

where $w_{q,q'}$ denotes a shortest separating word for the pair (q, q'). Hence every state defines a distinct residual sequential function, so no smaller deterministic subsequential transducer can realize the same map. As all ten states are reachable, the displayed transducer $\mathcal{K}$ itself is therefore a minimal realization. □

Corollary 6.4 (Exact state complexity)**.** *The classical Berstel adder has deterministic subsequential state complexity* 10*.*

Proof. This is exactly Theorem 6.3. □

Corollary 6.5. *The minimal deterministic subsequential realization of the function induced by $\mathcal{K}$ is unique up to labeled isomorphism.*

Proof. This is the standard uniqueness statement for minimal deterministic sequential machines; see [**?**]. □

7 A Local Parallel Lower Bound

The failure of bounded one-pass delay does not, by itself, rule out efficient synchronous local normalization. The next theorem shows that locality also fails in that model: the number of rounds must grow linearly with the input scale.

Theorem 7.1 (Local parallel lower bound for Zeckendorf normalization)**.** *Let $\mathcal{A} = \{0, 1, 2\}$ and define, for finitely supported one-sided configurations $x = (x_1, x_2, \ldots) \in \mathcal{A}^{(\mathbb{N})}$,*

$$\mathrm{Val}(x) := \sum_{k \geq 1} x_k F_{k+1}.$$

Let $Z(\mathrm{Val}(x)) \in \{0, 1\}^{(\mathbb{N})}$ denote the unique admissible Zeckendorf representative with the same value.

Suppose $\Phi : \mathcal{A}^{(\mathbb{N})} \to \{0, 1\}^{(\mathbb{N})}$ is a synchronous local map of radius r, meaning that after extending every input by zeros outside $\mathbb{N}$ there exists a local rule $f : \mathcal{A}^{2r+1} \to \{0, 1\}$ such that

$$(\Phi(x))_j = f(x_{j-r}, \ldots, x_{j+r}) \qquad (j \geq 1),$$

and let Φ^t denote its t-fold iterate. If for some $t \geq 1$ one has

$$\Phi^t(x) = Z(\mathrm{Val}(x))$$

for every finitely supported input x, then for every even integer $m \geq 2$,

$$t \geq \left\lceil \frac{m-1}{r} \right\rceil.$$

In particular, no fixed-radius synchronous local normalization scheme can run in a number of rounds independent of the input scale.

Proof. After t iterations, the output coordinate at position j depends only on the input in the interval $[j - rt, j + rt] \cap \mathbb{N}$. Fix an even integer $m \geq 2$ and define

$$x_k^{(m)} = \begin{cases} 1, & k \in \{1, 3, 5, \ldots, m-1\}, \\ 0, & \text{otherwise}, \end{cases} \qquad y^{(m)} := x^{(m)} + e_1.$$

The configuration $x^{(m)}$ is already admissible, hence $Z(\mathrm{Val}(x^{(m)})) = x^{(m)}$ and its mth coordinate is 0.

On the other hand,

$$\mathrm{Val}(x^{(m)}) = \sum_{j=1}^{m/2} F_{2j} = F_{m+1} - 1,$$

so

$$\mathrm{Val}(y^{(m)}) = \mathrm{Val}(x^{(m)}) + F_2 = F_{m+1}.$$

Therefore $Z(\mathrm{Val}(y^{(m)}))$ is the admissible word with a single 1 at position m, and hence

$$Z(\mathrm{Val}(x^{(m)}))_m = 0, \qquad Z(\mathrm{Val}(y^{(m)}))_m = 1.$$

If $rt < m - 1$, then the coordinate $(\Phi^t(\cdot))_m$ cannot depend on the input at position 1, so it must take the same value on $x^{(m)}$ and $y^{(m)}$. This contradicts the required normalization outputs above. Hence $rt \geq m - 1$, which is equivalent to the stated bound. □

Remark 7.2. Theorem 7.1 concerns the nonredundant input alphabet $\{0, 1, 2\}$ and the canonical Zeckendorf output map $x \mapsto Z(\mathrm{Val}(x))$. It does not contradict constant-time parallel addition results for redundant digit sets, where carries can be absorbed by extra local freedom. We also do not claim a matching upper bound here: the theorem only shows that every fixed-radius synchronous realization of the canonical map requires a number of rounds growing linearly with the input scale.

8 Conclusion

We have separated three layers of the Zeckendorf addition problem. First, the canonical normalization map Fold_∞ is the section of the Fibonacci congruence selected by admissible representatives, and it admits a direct rewrite-theoretic description. Second, in the one-sided low-to-high model, this canonical map has coordinatewise unbounded anticipation and therefore no bounded online delay; more strongly, its low-to-high finite-word incarnation is not subsequential. Third, the classical 10-state Berstel adder belongs to a different, most-significant-digit-first subsequential setting, and in that setting its state complexity is exactly 10. The local lower bound shows that canonical normalization also remains nonlocal in the synchronous setting. Natural next questions concern analogous distinctions for other Pisot numeration systems and for redundant or signed digit alphabets. Machine-checked verification of the rewrite system and of the minimal subsequential quotient of the classical kernel would also be worthwhile.